

M-BEV: Masked BEV Perception for Robust Autonomous Driving

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- Problem / objective
 - BEV Perception when one or more cameras are failed
- Contribution / Key idea
 - Propose a generic Masked BEV (M-BEV) perception framework
 - Develop a novel Masked View Reconstruction (MVR) module
 - Perform extensive experiments on the NuScenes benchmark

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Motivation

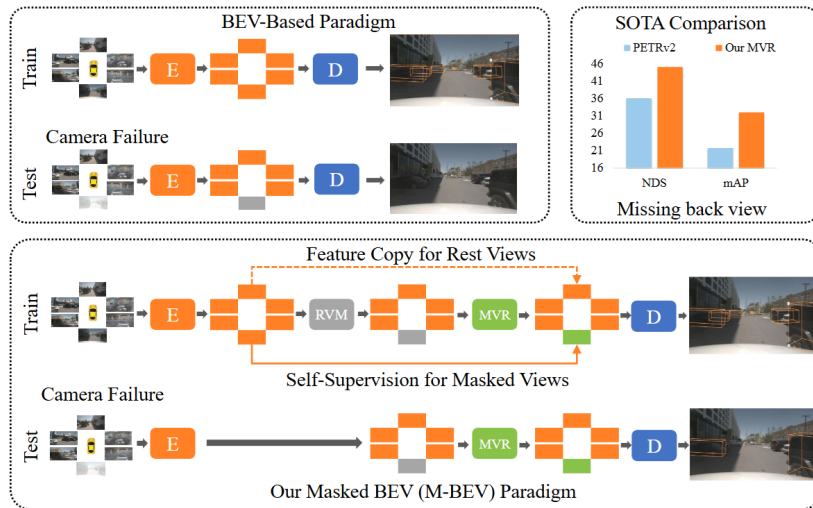


Figure 1: Motivation. In cases where cameras crash, the existing BEV-based approaches would be largely deteriorated. We design a self-supervised Masked View Reconstruction (MVR) module which can significantly boost the state-of-the-art models for various missing camera cases. E: Encoder, D: Decoder, RVM: Random View Masking.

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- Overview

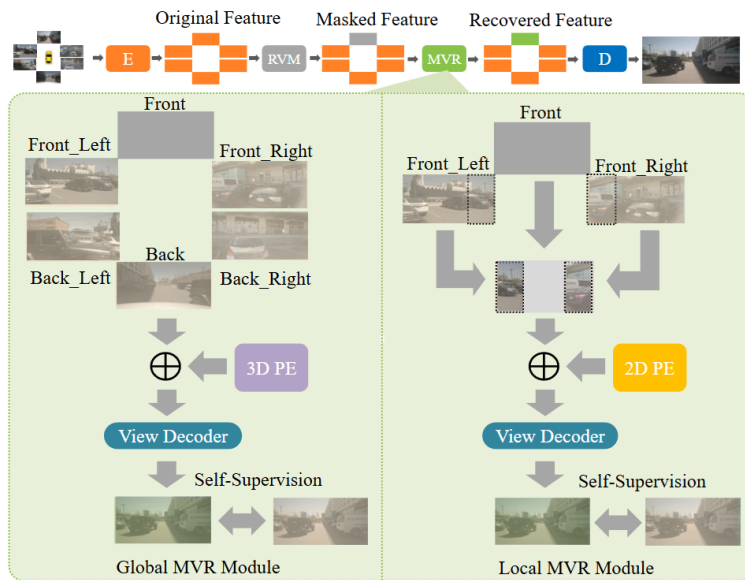


Figure 2: Overall Architecture of Our Proposed M-BEV Framework. M-BEV consists of 3D Object Detector with Visual Encoder (E) and Decoder (D), Random View Masking (RVM) and Masked View Reconstruction (MVR) modules. We propose two MVR modules, namely Global MVR and Local MVR, to recover the features of masked views.

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- Random View Masking (RVM)
 - 6개의 camera 중에 일부가 고장났거나 사용할 수 없는 상황을 가정하기 위한 모듈
 - 인코딩된 Multi-view image를 랜덤하게 선택해서 masking

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- Masked View Reconstruction (MVR)

• Global MVR

- masked view를 제외한 나머지 view 전체를 활용하여 reconstruction

$$\mathbf{U}_{mask} = \text{Decoder}(\mathcal{C}[\mathbf{V}_{mask}, \mathbf{F}_{rest}] + \mathbf{P}_{3D}),$$

• Local MVR

- masked view와 인접한 view만을 활용하여 reconstruction

$$\mathbf{V}_{mask} = \mathcal{C}[\mathbf{F}_{left}(right), \mathbf{V}_{mask}(mid), \mathbf{F}_{right}(left)].$$

$$\mathbf{U}_{mask} = \text{Decoder}(\mathbf{V}_{mask} + \mathbf{P}_{2D}),$$

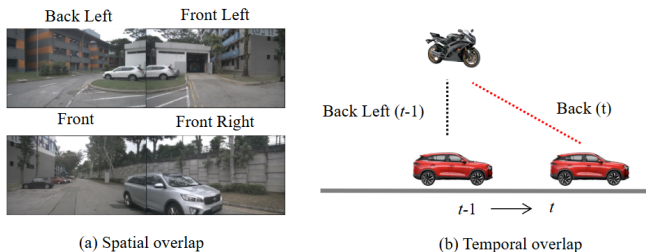


Figure 3: Illustration of spatial and temporal overlap.

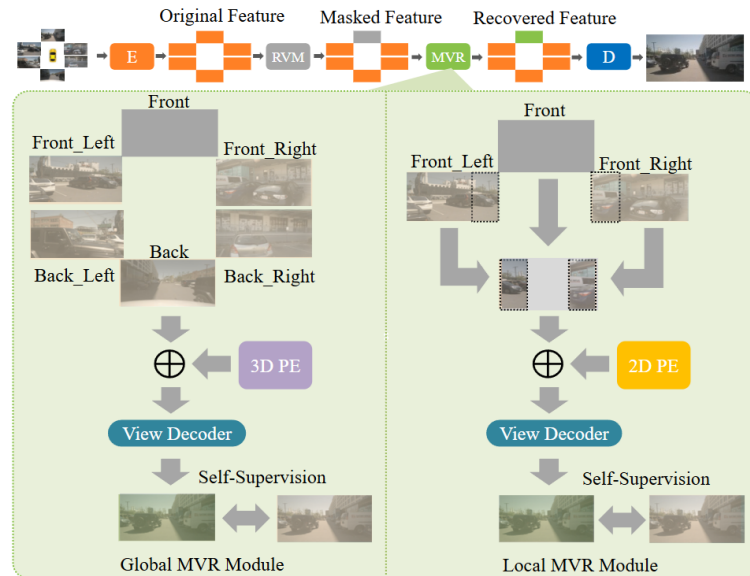


Figure 2: Overall Architecture of Our Proposed M-BEV Framework. M-BEV consists of 3D Object Detector with Visual Encoder (E) and Decoder (D), Random View Masking (RVM) and Masked View Reconstruction (MVR) modules. We propose two MVR modules, namely Global MVR and Local MVR, to recover the features of masked views.

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- Training

- original feature와 reconstructed feature 간에 L2 loss를 사용하여 decoder pre-training

$$\mathcal{L}_{mvr} = \|\mathbf{F}_{mask} - \mathbf{U}_{mask}\|_2.$$

- object classification과 3D bounding box regression을 위한 $\mathcal{L}_{det}$ 를 사용하여 detection head fine-tuning

$$\mathcal{L}_{total} = \mathcal{L}_{det} + \alpha \mathcal{L}_{mvr},$$

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- Result

W/O Missing	Method	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	MAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
Standard	PETrv2 (Liu et al. 2023)	0.4853	0.3977	0.7531	0.2693	0.4978	0.4310	0.1840
Missing	Method	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	MAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
Front	PETrv2 (Liu et al. 2023)	0.4238	0.3022	0.7757	0.2740	0.5311	0.5043	0.1883
	Our M-BEV (PETrv2)	0.4504	0.3263	0.7234	0.2736	0.4996	0.4449	0.1862
Front_Right	PETrv2 (Liu et al. 2023)	0.4363	0.3294	0.8444	0.2706	0.5333	0.4551	0.1808
	Our M-BEV (PETrv2)	0.4712	0.3666	0.7346	0.2704	0.5098	0.4214	0.1850
Front_Left	PETrv2 (Liu et al. 2023)	0.4405	0.3355	0.8195	0.2733	0.5216	0.4664	0.1912
	Our M-BEV (PETrv2)	0.4678	0.3628	0.7308	0.2740	0.5113	0.4298	0.1905
Back	PETrv2 (Liu et al. 2023)	0.3616	0.2179	1.0176	0.2977	0.5618	0.4477	0.1726
	Our M-BEV (PETrv2)	0.4516	0.3206	0.7283	0.2688	0.4908	0.4237	0.1754
Back_Left	PETrv2 (Liu et al. 2023)	0.4568	0.3513	0.7910	0.2700	0.4903	0.4476	0.1895
	Our M-BEV (PETrv2)	0.4753	0.3694	0.7277	0.2694	0.4770	0.4291	0.1909
Back_Right	PETrv2 (Liu et al. 2023)	0.4556	0.3544	0.7892	0.2700	0.5157	0.4508	0.1902
	Our M-BEV (PETrv2)	0.4756	0.3730	0.7294	0.2711	0.4990	0.4211	0.1879

Table 1: Performance comparison on PETrv2 (Liu et al. 2023) when losing each of six camera views. The effect of our M-BEV is impressive, e.g., for the absence of back view, our M-BEV achieves 10.3% mAP improvement.

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- Result

W/O Missing	Method	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
Standard	BEVStereo (Li et al. 2023)	0.4432	0.3439	0.6583	0.2823	0.5860	0.5287	0.2327
Missing	Method	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
Front	BEVStereo (Li et al. 2023)	0.3901	0.2462	0.6799	0.2867	0.6192	0.5159	0.2283
	Our M-BEV (BEVStereo)	0.4027	0.2667	0.6744	0.2866	0.6223	0.5103	0.2130
Front_Right	BEVStereo (Li et al. 2023)	0.4042	0.2832	0.6770	0.2841	0.6027	0.5811	0.2298
	Our M-BEV (BEVStereo)	0.4148	0.3004	0.6733	0.2845	0.6122	0.5791	0.2153
Front_Left	BEVStereo (Li et al. 2023)	0.4039	0.2795	0.6720	0.2820	0.5960	0.5645	0.2429
	Our M-BEV (BEVStereo)	0.4126	0.2967	0.6725	0.2851	0.6134	0.5690	0.2269
Back	BEVStereo (Li et al. 2023)	0.3894	0.2373	0.6761	0.2840	0.6071	0.5126	0.2126
	Our M-BEV (BEVStereo)	0.3949	0.2580	0.6703	0.2880	0.6196	0.5540	0.2087
Back_Left	BEVStereo (Li et al. 2023)	0.4132	0.2891	0.6753	0.2828	0.5687	0.5457	0.2408
	Our M-BEV (BEVStereo)	0.4149	0.2989	0.6762	0.2859	0.5966	0.5709	0.2259
Back_Right	BEVStereo (Li et al. 2023)	0.4081	0.2898	0.6813	0.2817	0.5945	0.5739	0.2367
	Our M-BEV (BEVStereo)	0.4157	0.3027	0.6719	0.2843	0.6052	0.5738	0.2209

Table 2: Performance comparison on BEVStereo (Li et al. 2023) when losing each of six camera views. Overall, our M-BEV gives a comprehensive improvement for the baseline.

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- Result

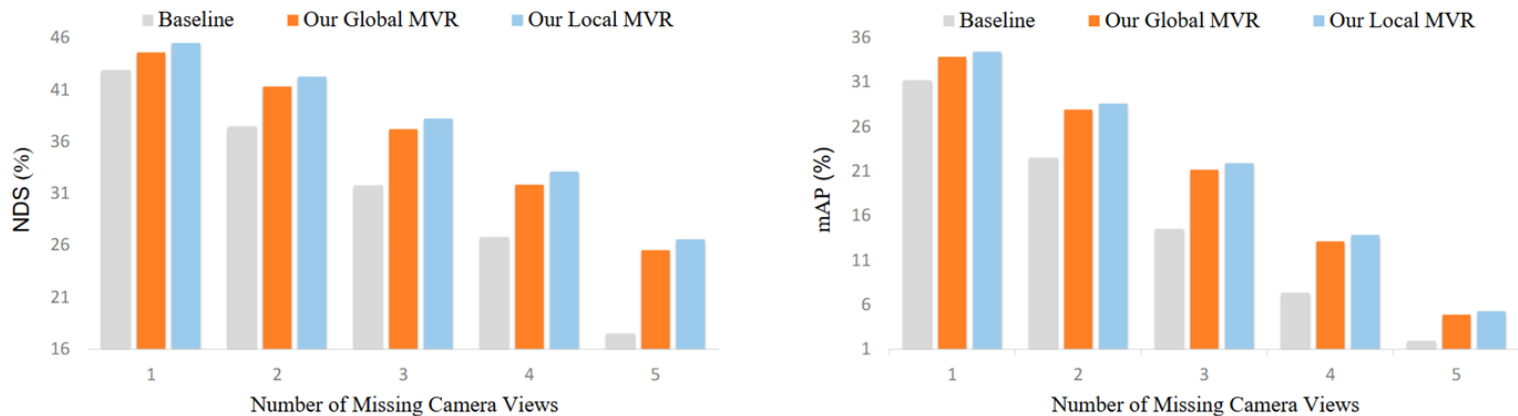


Figure 4: Performance of baseline, our Global MVR and Local MVR settings for different number of missing cameras.

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- Result

Missing	Our M-BEV	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	MAOE $\downarrow$	MAVE $\downarrow$	MAAE $\downarrow$
Front	Global MVR	0.4361	0.3160	0.7524	0.2725	0.5489	0.4584	0.1873
	Local MVR	0.4504	0.3263	0.7234	0.2736	0.4996	0.4449	0.1862
Front_Right	Global MVR	0.4588	0.3592	0.7622	0.2684	0.5550	0.4371	0.1852
	Local MVR	0.4712	0.3666	0.7346	0.2704	0.5098	0.4214	0.1850
Front_Left	Global MVR	0.4574	0.3570	0.7635	0.2727	0.5410	0.4410	0.1928
	Local MVR	0.4678	0.3628	0.7308	0.2740	0.5113	0.4298	0.1905
Back	Global MVR	0.4390	0.3146	0.7642	0.2679	0.5340	0.4400	0.1773
	Local MVR	0.4516	0.3206	0.7283	0.2688	0.4908	0.4237	0.1754
Back_Left	Global MVR	0.4632	0.3633	0.7645	0.2697	0.5214	0.4350	0.1944
	Local MVR	0.4753	0.3694	0.7277	0.2694	0.4770	0.4291	0.1909
Back_Right	Global MVR	0.4629	0.3671	0.7638	0.2694	0.5504	0.4333	0.1895
	Local MVR	0.4756	0.3730	0.7294	0.2711	0.4990	0.4211	0.1879

Table 3: The performance after reconstruction of two MVR module variants. Local MVR is a better choice by exploiting the distinct context from adjacent camera views.

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MVR	FineTune	PE	MaskRatio	NDS $\uparrow$	mAP $\uparrow$	mATE $\downarrow$	mASE $\downarrow$	mAOE $\downarrow$	mAVE $\downarrow$	mAAE $\downarrow$
$\times$	$\times$	$\times$	$\times$	0.4285	0.3117	0.8279	0.2730	0.5228	0.4638	0.1865
$\times$	$\checkmark$	$\times$	$\times$	0.4466	0.3371	0.7752	0.2741	0.5256	0.4469	0.1975
$\checkmark$	$\times$	$\checkmark$	76%	0.4319	0.3130	0.8196	0.2721	0.5181	0.4545	0.1852
$\checkmark$	$\checkmark$	$\times$	76%	0.4552	0.3477	0.7556	0.2715	0.5190	0.4376	0.1870
$\checkmark$	$\checkmark$	$\checkmark$	76%	0.4580	0.3499	0.7544	0.2712	0.5186	0.4368	0.1878
$\checkmark$	$\checkmark$	$\checkmark$	60%	0.4576	0.3495	0.7538	0.2710	0.5171	0.4402	0.1885
$\checkmark$	$\checkmark$	$\checkmark$	64%	0.4566	0.3474	0.7529	0.2720	0.5185	0.4378	0.1894
$\checkmark$	$\checkmark$	$\checkmark$	68%	0.4570	0.3491	0.7596	0.2714	0.5246	0.4302	0.1892
$\checkmark$	$\checkmark$	$\checkmark$	72%	0.4579	0.3485	0.7540	0.2721	0.5126	0.4352	0.1895
$\checkmark$	$\checkmark$	$\checkmark$	80%	0.4560	0.3478	0.7578	0.2716	0.5249	0.4330	0.1916

Table 4: The ablation of MVR module, fine-tuning, position embedding and masking ratio for M-BEV. All models are trained with same schedule and hyper-parameters. We randomly drop one camera and calculate the average metrics for inference.

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- Result

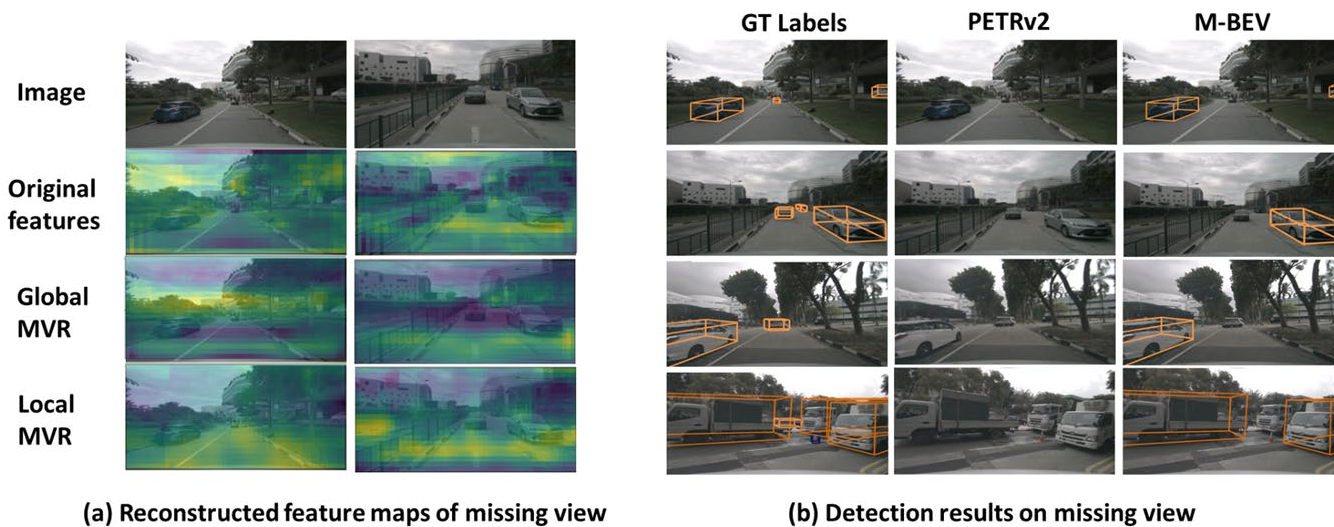


Figure 5: Visualization for feature maps and detection results. M-BEV reconstructed features could be a rough substitute of the original feature, and could help for the detection of the vehicles on the left and right sides of the missing view.

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W/O Missing	NDS↑	mAP↑	GFLOPS↓	Time(ms)↓
PETrv2 Baseline	0.4853	0.3977	1047	—
Our Global MVR	0.4872	0.4007	1047	+0.0
Our Local MVR	0.4898	0.4039	1047	+0.0
Missing	NDS↑	mAP↑	GFLOPS↓	Time(ms)↓
PETrv2 Baseline	0.4285	0.3117	1047	—
Our Global MVR	0.4534	0.3467	1088	+1.94
Our Local MVR	0.4580	0.3499	1053	+0.61

Table 5: Generalization vs. Computation. The model trained with our MVR performs better the original baseline under both no-missing and missing settings, while requiring only little extra GFLOPS for computation.